Transparent Arithmetic in Combinators: Specification and Design Space

Concept note v1.0, Project secret2, August 10, 2026

1. Goal

Project secret1 built a number system on {K, S} trees and characterized primality twice: as universal indecomposability under grafting, and as unreachability of product boxes under an inverse rewriting system [1]. Both characterizations re-express trial division, and the unary comb encoding makes every divisibility probe cost linear in n.

Project secret2 raises the bar. We want a combinator representation of the natural numbers in which testing primality and extracting factors is immediate: visible in the structure of the term, at cost far below re-deriving it. This note fixes what 'immediate' can coherently mean, specifies the properties such a representation must balance, maps the known design points, and sets the experimental program. Experiments E1 (binary numerals from doubling and successor), E2 (divisibility as local rewrite automata), E3 with its continuation E3.2 (F_arith, the factoring orientation, then binary leaves with rewrite division), E4 (the enumerative search, exhaustive over all weak normal forms to fifteen leaves and all raw terms to nine), and E5 (the residue bundles) are complete: the full experimental program of this note is delivered, and Section 6 reports all results.

2. What 'immediate' can and cannot mean

Three external facts bound the search space before any design work starts.

2.1 The conversion barrier

Observation 1. Suppose a representation R comes with: encoding from standard binary in time polynomial in log n; decoding back in time polynomial in the representation size; representation size polynomial in log n; and a factor-extraction operation that, for composite n, produces representations of some a, b with n = ab and a, b at least 2, in time polynomial in the representation size. Then integer factorization is polynomial-time: encode, extract, decode, and recurse, with recursion depth at most log n.

No such algorithm is known, and much of cryptography wagers that none exists. So every honest design drops at least one leg, and the choice of which leg to drop is the design decision:

- Drop cheap encoding: the representation is the factored form itself. Entry costs a factorization; afterward multiplication, divisibility, primality, and factor extraction are literally free, while addition becomes as hard as factoring the sum.
- Drop compactness: unary. Project secret1 lives here; its full primality certificates cost order n squared steps, exponential in the input length, exactly as the barrier predicts.
- Drop full extraction: residue systems expose only factors below a smoothness bound; everything above it is invisible.
- Drop nothing, but relativize: make transparency a property of canonical forms under an equality whose normalization carries the cost. This is the move native to this project, and Section 2.4 turns it into the working definition.

2.2 Testing is not extraction

Primality testing is in polynomial time unconditionally [5]. So a compact representation whose canonical forms wear a structural primality marker contradicts no complexity assumption; only fac-

tor extraction is barred by Observation 1. The two goals separate, and the specification below keeps them separate (P6 versus P7).

2.3 The locality obstruction for positional encodings

For digit-style representations the obstruction to primality is not complexity but locality. The binary primes are not a regular language [2]; neither the primes nor any infinite subset of them is context-free in any base [3]; and any finite automaton that is merely correct on inputs up to N needs a number of states polynomial in N [4]. Consequence: if R(n) is, up to renaming, the base-b digit word of n, then no bounded root window, no finite tree automaton, and no approximately correct small automaton decides primality on it. Structural primality forces the representation away from positional encodings altogether, toward factored forms, residue bundles, or something not yet on the map.

2.4 The working definition

Let =E be an equality on terms generated by a rewriting system with unique normal forms on the fragment of interest. Call R **transparent for a property Q under =E** if the =E-normal form of R(n) decides Q within a root window of bounded depth, by a fixed pattern that does not grow with n.

The cost of Q then lives in normalization, not in the representation, and the research question becomes concrete: which equality has factored forms as its normal forms, how expensive is normalization, and where exactly does the exponential hide? Under this definition the fundamental theorem of arithmetic itself becomes a rewriting statement: normal forms are factored forms, and their uniqueness is the confluence of the factoring system. Project secret1 already listed that formulation as open; experiment E3 now realizes it over unary prime leaves, and Section 6 reports the results.

3. Specification: the property schema

The schema doubles as the fitness function for the enumerative search (E4). For a candidate representation R with ambient equality =E:

- **P0 adequacy.** R is injective on values; zero or one and a finite constructor set are representable; canonical forms are constructible for every n.
- **P1 size.** Tree size and DAG size (distinct subterms) of R(n), as functions of n. E1 shows the two can differ exponentially.
- **P2 canonicity.** Each number has a unique =E-normal form, and =E is decidable on the fragment.
- **P3 arithmetic.** Combinators for successor, doubling, addition, and multiplication, with normalization cost bounds relative to representation size.
- **P4 order.** Zero test and comparison, with costs.
- **P5 divisibility transparency.** For each k, the window depth d(k) such that n mod k is a function of the depth-d(k) root window of the canonical form, or else the scan cost when no bounded window suffices.
- **P6 primality transparency.** A structural predicate, a fixed tree-automaton pattern on canonical forms, deciding primality. Carried annotations do not count; the shape itself must speak.
- **P7 factor extraction.** A rewrite or combinator that reaches representations of a nontrivial factor pair from R(n), with cost profile.
- **P8 conversion.** Encoding from and decoding to standard binary, with costs. Observation 1 links P1, P3, P7, and P8: they cannot all be cheap.
- **P9 closure.** Transparency (P5 to P7) is preserved by the operations of P3; a marker that dies after one addition is decoration, not structure.

4. The design space, mapped

Representation	Tree size	DAG size	Add	Mult	Div by 2^j	Div by k	Primality	Factor extraction	Encode
Unary combs [1]	n	n	concat	R_arith, n k steps	scan n	scan n	aperiodicity, scan	stride search, n^2	n steps
Church numerals	n	n	easy	easy	opaque	opaque	opaque	opaque	n steps
Binary, doubling and successor (E1)	n	log n	shift-add	shift-add	window depth j	digit scan, log n	blocked by 2.3	probes plus division, log per split (E3.2)	log n
Scott-style BinPos	log n	log n	programmed	programmed	window depth j	digit scan, log n	blocked by 2.3	open	log n
Factored forms	log n	log n	as hard as factoring	multiset union, free	free	free	free (singleton test)	free	factoring
Residue bundles (CRT)	log n	log n	coordinate-wise	coordinate-wise	local per bundled prime	local if k bundled	smoothness only	small factors only	log n per prime
Hereditarily binary (Tarau) [8]	log n	log n	structural	structural	partial	opaque	opaque	opaque	log n

Reading the table against Observation 1: the factored-form row is the only one with free extraction, and it pays at the addition and encoding columns, exactly as the barrier demands. The residue row buys locality for every bundled prime and pays with partiality and with hard comparison. The positional rows are blocked at the primality column by Section 2.3 no matter how the arithmetic is tuned. The unary row is the control group: everything visible, everything expensive.

5. Reading K and S arithmetically

The philosophical route starts from what the two combinators do. K x y keeps its first argument and discards the second: separation, projection, hiding. S a b z feeds one input to two consumers: fusion, sharing, duplication.

Project secret1 already runs on this reading without naming it. In the product box S(K m) k, the S at the root fuses the incoming argument z into both branches, and the K shields m from it, so m receives k z rather than z. Composition is exactly 'share the input, hide it from the outer factor'; the frozen boxes are K-guarded potential energy, and R_arith spends it through the S-duplication in

rule R1, which is also why the termination measure fails. E1 adds a positive face of the same coin: S-sharing is why binary numerals are DAG-compact.

Two predictions follow. First, any design that K-guards a primality marker at the root must pay for maintaining it elsewhere, by Observation 1; there is no free flag. Second, and constructively: the S-as-fusion reading points at residue bundles. Represent n as an S-fused bundle of cyclic counters, one per small prime p, each counter a comb of length n mod p; K-projections read out a coordinate. Addition and multiplication act coordinatewise and locally; divisibility by every bundled prime is a root window on its coordinate; primality up to the smoothness bound is a conjunction of nonzero tests. The costs, hard comparison and blindness above the bound, are the ones the barrier forces anyway. Residue bundles are the K/S-native number system, and experiment E5 will build them.

6. Program of experiments

E1, done: binary numerals from doubling and successor. Numerals are generated by one = SB(KI), doubling dbl m = S(S(KB)m)m (the self-sum box of secret1), and successor suc m = SB m; the canonical form of n follows its binary expansion. Machine-checked results (Secret2.idr, Idris2):

- Adequacy: bin n and the unary numeral agree pointwise under weak reduction for n = 1 to 24.
- Parity is root-local for n = 1 to 64: the root window alone decides n mod 2 (doubling box versus successor link). Divisibility by 4 is depth-2-local, and by induction divisibility by 2^j is depth-j-local. This is the first genuinely immediate divisibility in the project.
- Compactness lives in the DAG, not the tree: bin 1024 has 16,377 leaves as a tree but only 42 distinct subterms; bin 255 has 2,676 leaves and 40 distinct subterms. Doubling duplicates syntactically; sharing restores log-size.
- The frozen product persists: weak reduction sends B (bin 3)(bin 5) to S(K bin 3)(bin 5). The box phenomenon is representation-independent.
- The doubling combinator exists inside the system: bracket abstraction of the term for 'apply twice' yields D = S(S(KS)(S(K(S(KS)K))))(SKK), and D (bin 3) weak-normalizes to exactly bin 6.
- Orientation conflict: R_arith flattens the structure it should preserve; its normal form of bin n is the unary comb (verified for 6, 7, and 12). The equality of secret1 points away from compact forms. Paper 2 therefore needs its own equality, oriented toward structure.

E2, done: divisibility automata as local rewrite rules. For each divisor k there is a finite family of local rules over states (r, p) in Z_k by Z_k: a probe term S(K(S(S(K r)(K p))) t carries the accumulated residue r and the current bit weight p over an untouched binary numeral t, and three rule schemata walk the digit word from the least significant bit inward. Mod0 crosses a doubling box (p doubles mod k), Mod1 crosses a successor link (r gains p, then p doubles), and ModC hits the terminator and emits the residue comb of (r + p) mod k. Machine-checked results (Secret2.idr, Idris2):

- Soundness: every applied rule preserves the affine invariant (r + p times value) mod k, checked at each step, and the emitted residue equals n mod k for all n = 2 to 40 and k = 2 to n.
- Cost: every probe takes exactly the binary length of n in steps, independent of k. The residue vectors of secret1 reproduce identically, at 12 against 35 steps for n = 6, 44 against 138 for n = 13, and 52 against 198 for n = 15.
- Primality certificates, probes for k = 2 up to the integer square root of n: 97 is prime in 56 local rewrite steps, 1,009 in 300, and 8,191 in 1,157. Composites emit their smallest factor as the first zero residue: 91 gives 7, 143 gives 11, and 1,001 gives 7. Small-factor extraction is free at certificate time.
- The dead-end anatomy of secret1 transforms as predicted: the residue is no longer a stranding shape reached after order n steps but the terminal state of an automaton, emitted after logarithmically many.

- The probe wrapper is inert under weak reduction and introduces no R_arith redex; R_arith still flattens the binary payload itself, confirming again that the ambient equality of secret1 opposes this representation.
- Consistency with Section 2.3: each fixed k is a finite automaton, and primality needs the unbounded family of probes up to the square root; no single finite rule set can replace the family.

E3, done: F_arith, the factoring orientation. The system rewrites toward factored forms: product boxes S(K a)b over numeral leaves. One certified rule family splits a composite comb leaf v into S(K k)((v/k) as a comb) for any proper divisor k, and every applied instance is certified by normalizing the box forward under R_arith and comparing with the leaf it replaced. Three canonicalization rules, Fassoc, Fsort, and Fswap, rotate boxes to the right and sort leaves ascending; each is pointwise valid, machine-checked. Results (Secret2.idr, Idris2):

- Primality is F-irreducibility. For n = 2 to 200, the numeral of n admits no split move exactly when n is prime: a prime is a term on which the factoring system has nothing to say.
- Termination is provable, in contrast with the open strong normalization of R_arith: splits strictly decrease the multiset of leaf values, and the canonicalization rules decrease left depth and inversion count.
- Confluence is unique factorization. Every first split of every composite n up to 100 rejoins at the canonical ascending prime form (n = 60 alone has 10 ordered first splits), and full derivations with randomized split choices over three seeds reach the same normal form. The fundamental theorem of arithmetic appears as the joinability of split peaks.
- The probe-guided pipeline factors with certified steps: 30 into 2, 3, 5 in two splits (13 probe steps, 57 certification steps); 143 into 11, 13 in one split (80 and 166); 210 into 2, 3, 5, 7 in three splits (46 and 373); 1,001 into 7, 11, 13 in two splits (140 and 1,182); 1,024 into ten twos in nine splits (63 and 2,089). The prime 97 is irreducible at zero splits.
- Cost anatomy, exactly what the experiment was for: discovery is cheap and certification is expensive. For 1,001 the probes cost 140 steps while the forward certifications cost 1,182, and each unary certification of a split costs order v steps for a leaf of value v. Observation 1 has been localized: the residual cost sits in certifying splits over unary leaves. E3.2, the planned continuation, replaces unary leaves and forward certification by binary leaves and rewrite division, aiming at polylogarithmic certificates per split. It is delivered directly below.

E3.2, done: binary leaves and rewrite division. The split certificates of E3 move from unary forward multiplication to division rewrites on binary leaves. Division by 2 is one root-local step: the doubling box at the root of an even leaf is itself the certificate, parity locality cashed in. Division by an odd prime k runs least-significant-bit first with a bounded state r < k, a Hensel-style transition family reusing the E2 marker shape, one transition per binary digit, each checked against the all-natural-number invariant k times Q plus 2^i times rest equals v plus 2^i times r; a nonzero final state refuses non-divisors. Results (Secret2.idr, Idris2):

- Certificate length per split is the binary length of the leaf, and the quotient term is produced by the run itself, matching the canonical numeral exactly.
- The cost collapse across the pipeline: 30 drops from 57 unary certification steps to five division steps, 1,001 from 1,182 to 18, 1,024 from 2,089 to nine (each halving is one root-local step), and 4,095 from 6,046 to 39. For 65,535, which factors as 3, 5, 17, 257 and where unary certification is impractical, the full certified factorization costs 300 probe steps and 44 division steps.
- The Fermat prime 65,537 is certified prime by the probe family in 4,335 steps.
- With this, every ingredient of the factoring equality costs one step per binary digit except one: discovery still probes candidate divisors in ascending order, so the square-root wall of trial division remains. That is exactly where Observation 1 says the irreducible cost must live for a representation with cheap encoding.

E4, frontier done to fifteen leaves. The enumeration now covers all 278,707,802 closed weak normal forms up to fifteen leaves, plus all raw terms up to nine, under a uniform screen. Three ingredients made the scale feasible. Two pruning lemmas: every K-headed constructor is a constant map, S K x is the identity for every x, and S S x is never an iterator (its evaluation stacks the iteration variable over a residual argument); the pruned classes were also scanned empirically at full scale and kept exactly zero candidates. A Krivine-style weak head machine with an argument stack replaces the root-rescanning reducer in the scan path, one unit of fuel per rule firing, with the iterator value read off the head spine; the original reducer remains in the rank stage as an independent cross-check. And an engineering repair: the standard library concatenation is left-folded and turns enumeration quadratic near 10^5-element outer lists; a right-folded concatenation restored linearity and turned a candidate class that hung for minutes into a one-second scan. Results (Secret2.idr, entry points e4scan with part modes and e4rank):

- The census: 656 keepers reduce to 33 distinct orbit signatures (a signature is an orbit prefix, so budget-truncated duplicates of one map can appear twice; at least 24 distinct arithmetic actions remain after identifying prefixes). Beyond the five known at eleven leaves, the twelve-leaf stratum alone contributes the additive step n + 2, the honest 2n + 1 with orbit 1, 3, 7, 15 (the eta-safe form of the map whose naive S(SB) is weakly stuck), tripling with orbit 1, 3, 9, 27, the square (n + 1) squared, and the exponential 2 to the n with orbit 1, 2, 4, 16; thirteen leaves adds, among others, a map with orbit 1, 2, 16, consistent with 2 to the n squared; and fourteen adds quadrupling with orbit 1, 4, 16, 64, the shifted exponential 2 to the (n + 1) with orbit 1, 4, 32, and the map 2 n squared with orbit 1, 2, 8, 128; fifteen adds, among others, the map n(2n + 1) with orbit 1, 3, 21. The pruned classes, 71.7 million candidates at fifteen alone, kept exactly zero at every size where they were scanned.
- The null result: 781 alphabets arise over the 40 smallest actions; 104 cover the window 2 to 10 with 5 and 6 present. Exactly one is promising at stage 1, a pair of 2n + 1 with a doubling variant separating primes by normal form size, and stage 2 eliminates it on 2 to 30: the second finite-window artifact the staged screen has promoted and executed. Nothing survives.
- Next: the pool at sixteen leaves, about 0.95 billion normal forms, and the binary composition grammar, which provably rediscovers the E3 boxes.

E5, done: residue bundles. The K/S-native design of Section 5, built and measured. A bundle is a right-nested chain of inert two-slot carriers, pair a b = S(S(K a))(K b), one cyclic residue comb per basis prime; for the basis 2, 3, 5, 7 the term has constant size (78 leaves for 97) and is weak and R_arith normal. The reading of Section 5 is a reduction fact: pair a b X reduces to a (X b), S fusing the argument into both branches while each K shields its residue. Machine-checked: divisibility by the i-th basis prime is a window read at depth i for all n below 210 and for the six-prime basis with modulus 30,030; primality is a conjunction of window reads on the validity windows below 121 and 289, verified exhaustively; addition and multiplication are coordinatewise rule tables costing exactly one local rewrite per coordinate, four steps on the four-prime basis, independent of the operands, verified over all 3,600 operand pairs below 60; the encoder composes from the E2 probes, assembling the bundle of 8,191 in 78 local steps; and the paid costs arrive on schedule, the successor iterated 211 times from zero returning to the bundle of 1, the product of the bundles of 6 and 35 being the zero bundle, and comparison admitting no bounded window. The design point the table of Section 4 predicted is now occupied, at exactly the predicted price.

7. References

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